MIL-S-6790F(USAF)
23 December 1971
SUPERSEDING
MIL-S-6790E(USAF)
7 September 1965

MILITARY SPECIFICATION

SUSPENDERS, TROUSERS
(SUSPENDERS, FLYING SUIT)

1. SCOPE

1.1 This specification covers suspenders for use with flying clothing.

1.2 Classification. The suspenders shall be furnished in the following types as specified (see 6.2):

Type I - Elastic

Type II - Nonelastic

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton
QQ-S-630	Steel, Bar, Carbon, Hot Rolled (Merchant Quality)
QQ-S-698	Steel, Sheet And Strip, Low Carbon
UU-P-268	Paper, Kraft, Untreated, Wrapping
DDD-L-20	Label; For Clothing, Equipage, And Tentage, (General Use)
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape: Pressure - Sensitive Adhesive; Waterproof, For Packaging

Military

MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural Or In Colors
MIL-L-3754	Leather, Cattlehide, Chrome-Tanned, Strap, Pad, And Reinforcement
MIL-W-5664	Webbing, Textile, Elastic Cotton
MIL-F-10884	Fasteners, Snap

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STANDARDS

Federal

FED-STD-151	Metals, Test Methods
FED-STD-751	Stitches, Seams, And Stitchings

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Guide samples. Guide samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variation from this specification may appear in the sample in which case this specification shall govern.

3.1.1 First article approval. The requirement for the first article will be as specified by the Procuring Activity (see 6.2).

3.2 Materials.

3.2.1 Webbing. The webbing shall consist of two strips, 1-1/2 inch wide. The color of the webbing shall be sage green USAF color shade No. 1514 (see 6.3).

3.2.1.1 Elastic. The elastic webbing for the type I suspenders shall conform to class 2 of MIL-W-5664 and shall be cut into 34-inch long strips.

3.2.1.2 Nonelastic. The nonelastic webbing for the type II suspenders shall conform to type IIa of MIL-W-530 and shall be cut into 40-inch long strips.

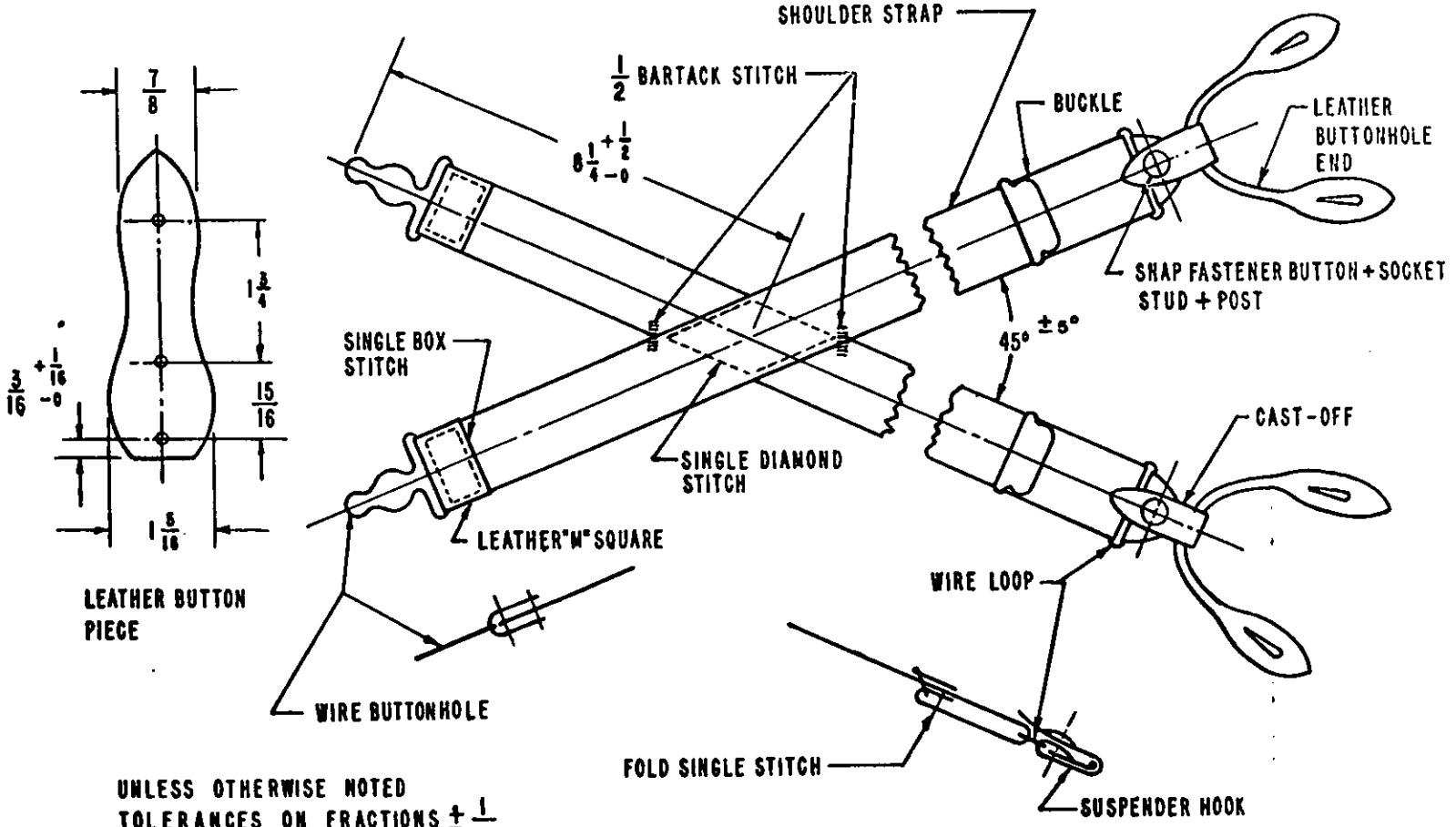
3.2.2 Thread. The thread for all sewing operations shall conform to type IA3, 4 ply, ticket No. 24 of V-T-276. The thread shall be vat dyed to match the color of the webbing.

3.2.3 Leather. The leather used in the construction of the suspenders shall be 3/64 to 4/64 inch in thickness and shall conform to type II of MIL-L-3754. The color of the leather shall be black or russet.

3.2.4 Hardware. The hardware shall conform to Figure 1 and shall be zinc plated and lacquered black.

* 3.2.4.1 Buckle. The front of the buckle shall be made from 0.020-inch, 3-temper, 1010 cold-rolled steel, conforming to QQ-S-698. The back of the buckle shall be made from 0.058-inch soft steel wire conforming to QQ-S-630. (See Table I, Note 2.)

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UNLESS OTHERWISE NOTED
TOLERANCES ON FRACTIONS ± 1/32
AND DIMENSIONS ARE IN INCHES

FIGURE 1 - SUSPENDERS

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* 3.2.4.2 Wire buttonholes. The wire buttonholes shall be made from 0.088-inch 1010 hot-rolled steel wire conforming to QQ-S-630, and the butted ends shall be welded together.

* 3.2.4.3 Castoff. The metal parts of the castoff shall consist of a wire loop made from 0.080-inch soft basic steel wire; a ferrule and a suspender hook made from 0.015-inch 3-temper 1010 hot-rolled steel conforming to QQ-S-630; and a snap fastener. The snap fastener shall consist of a button 20 line size, a socket, a stud, and an eyelet of sufficient length to provide for a firm clinch, conforming to style 2, finish 2 of MIL-F-10884.

3.3 Construction. The suspenders shall be constructed in accordance with Figure 1.

3.3.1 Castoff assembly. A leather button piece, 3-7/8 inches (plus or minus 1/8 inch) long, shall be cut in a shape similar to that shown on Figure 1. It shall be perforated with three holes so that, when folded over the wire loop, two of the holes line up to receive an eyelet, a metal suspender hook, and a stud. The hole in the tab end shall receive a button and a socket. The castoff, when snapped together, shall form a snug loop around the metal suspender hook and shall measure 1-1/2 inches (plus or minus 1/4 inch) in length.

3.3.2 Leather buttonhole. The leather buttonhole shall be die-cut from a piece of leather 8-1/4 inches (plus or minus 1/8 inch) long by 1 inch (plus or minus 1/8 inch) wide. A 7/8 inch long buttonhole shall be cut 1/2 inch from each end. The center portion of the leather shall be folded over with the cut edges even and shall be sewn for a distance of 4-1/2 inches (plus or minus 1/4 inch) in length with a single row of stitching 1/16 inch from the edges, forming a rounded finish.

* 3.3.3 Stitches, seams, and stitchings. All stitches seams and stitchings, except bartacking, shall be performed using stitch type 301 with 7 to 10 stitches per inch and shall conform to FED-STD-751.

* 3.3.3.1 Bartacking. Bartacking shall be 1/2 inch ±1/8 inch in length, 1/8 ±1/32 inch in width and shall contain a minimum of 28 stitches per tack. Bartacks shall be free from thread breaks, loose and skipped stitching.

* 3.3.3.2 Thread breaks, thread tensions and thread ends. Thread breaks and two or more consecutive skipped stitches or runoffs shall be overstitched not less than 1/2 inch in each direction beyond the defective stitching area. Ends of all stitching shall be backstitched or overstitched 3/4 inch minimum except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so that there will be no loose stitching resulting in a loose bobbin or top thread or excessively tight stitching resulting in puckering of material sewn. All ends shall be trimmed to 1/4 inch maximum length.

* 3.3.3.3 Setting of the metallic snap fasteners. Holes punched before inserting the male or female part of the fastener shall be smaller than the outside diameter of the fastener tubes so that the tubes must be forced through the holes. The fastener shall be securely clinched without cutting the leather.

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* 3.4 Assembly. One-half inch of the webbing shall be looped through the wire part of the buckle and single stitched 1/8 inch from the edges. The free end of the webbing shall then be looped through the wire loop of the castoff and then passed back through the buckle. The free end of the webbing shall be attached to the wire buttonhole by a die-cut leather "M" square, 2-1/2 inches (plus 1/8, minus 0 inch) long by 1-1/2 inches (plus 1/16 inch, minus 0 inch) wide. The leather "M" square shall be centered on the accommodating part of the buttonhole, folded over, and secured to the webbing with a single row of stitches 1/16 inch from the edge, forming a box 7/8 inch by 1-3/8 inches (plus or minus 1/16 inch). The webbing need not be caught in the stitching at the end of the wire. The two straps shall be crossed so the left shoulder strap will finish on top of the right and shall be single stitched to form a diamond back, 1/16 inch inside the edge of the webbing. The top and bottom of the diamond back shall be bartacked as shown in Figure 1.

3.5 Performance.

3.5.1 Salt spray. All metal parts used in the construction of the suspenders shall withstand a 100-hour exposure to salt spray without corrosion of the base metal.

* 3.6 Identification of product. One strap of each pair of suspenders shall be marked for identification in accordance with type IV, class 5 of DDD-L-20. The marking shall be in a contrasting color and shall include the following specific information:

Suspenders, Trousers
Contractor's Name 1/
Type
Contract No. 1/
Stock No. 1/

1/ The manufacturer shall include the applicable information.

3.7 Workmanship. The suspenders shall be constructed in a thoroughly workmanlike manner. The finished suspenders shall be clean and free from any defects that may affect appearance or functionability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of certification.

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* 4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Testing of components. Unless otherwise specified by the procuring activity, components shall be tested as specified in Table I.

4.2.1.1 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. In-process inspection to determine conformance to the requirements of 3.2.1.1 and 3.2.1.2 shall consist of inspection of the dimension of the cut length of webbing (elastic and non-elastic). In-process inspection to determine conformance to the requirements of 3.2.3 shall consist of inspection of the thickness of the leather button piece and leather "M" square prior to their attachment to the suspenders.

4.2.2 Examination of end item. The end item shall be examined for defects. Defects found during this examination shall be classified as specified in 4.2.2.1 and 4.2.2.2. The lots shall be inspected as specified in 4.2.2.3. The sample unit for this examination shall be one completely fabricated pair of suspenders.

4.2.2.1 General defects. General defects found during examination of the end item shall be classified as follows:

Defect	Major	Minor
1. WEBBING		
a. Holes, cuts, or tears	X	
b. Frayed or scalloped edges or not firmly and tightly woven	X	
c. Other webbing defects such as abrasive mark, slub, smash, or broken end or pick		
-seriously affecting serviceability or appearance	X	
-affecting serviceability or appearance but not seriously		X
2. LEATHER		
a. Holes, brand marks, or excessive softness	X	
b. Deep scratches or broken grains	X	
c. Color of all leather components not uniform		X
3. HARDWARE		
a. Broken or malformed, finish omitted, corroded areas, burr or sharp edge that might cause injury when handling	X	

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Defect	Major	Minor
4. SEAM AND STITCHING		
a. Open seams. (A seam shall be classified as an open seam when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-off occur.)		
-3/8 inch or less		X
-more than 3/8 inch	X	
b. Run-offs (see open seams)		
c. Raw edges not securely caught in stitching (Raw edges securely caught in stitching shall be classified as open seams.)		
(1) Any raw edge except at end of buckle		X
d. Seam and stitch type		
(1) Not specified seam or stitch type		
-seriously affecting serviceability or appearance	X	
-affecting serviceability or appearance but not seriously		X
e. Stitch tension		
(1) Loose stitch tension resulting in a loose seam or tight stitch tension resulting in puckering or excessively tight seams (stitches break when normal stress is applied)	X	
(2) Loose stitch tension resulting in a loosely exposed top or bobbin thread		X
f. Stitches per inch		
-one or two stitches less than minimum specified		X
-three or more stitches less than minimum specified	X	
-one or more stitches in excess of maximum specified		X
g. Stitching margins		
(1) Not as specified, affecting serviceability		X
h. Stitching ends		
(1) Not backstitched		X
i. Thread breaks. (Thread breaks not backstitched shall be classified as open seams.)		
j. Bartacks		
(1) One or both omitted, incomplete, or loose	X	
(2) Length or bite not as specified		X

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Defect	Major	Minor
(3) Misplaced -seriously affecting serviceability	X	
-affecting serviceability but not seriously		X
5. CLEANNESS		
a. Grease or oil stains clearly noticeable or thread ends not trimmed throughout		X
6. MARKING		
a. Omitted, incorrect, illegible, or not in contrasting color		X
7. COMPONENTS AND ASSEMBLY		
a. Any required detail of construction omitted or not as specified -seriously affecting serviceability or appearance	X	
-affecting serviceability or appearance but not seriously		X
b. Any required component omitted	X	
8. CONSTRUCTION DETAILS		
a. Cast-off assembly		
(1) Any part of cast-off assembly (wire loop or ferrule suspender hook) insecure or incomplete, affecting function of cast off or preventing leather buttonhole from engaging metal hook	X	
(2) Construction or shape of leather button pieces not as specified		X
(3) Reversed, i.e., not on straps with cast-off opening finished on the back side of buckle instead of front side	X	
(4) Stud or socket portion of snap fastener set off center of leather button piece, affecting serviceability or appearance		X
(5) Stud or socket not securely clinched so that either portion of snap fastener can become disengaged from leather button piece or clinched excessively tight, cutting the leather	X	

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Defect	Major	Minor
(6) Stud or socket not functioning properly, failing to effect a smooth and secure closure	X	
(7) Splits in roll of cap tube		X
(8) Button and socket of female or stud and eyelet of male snap fastener separate (The male and the female portions of the fastener shall be tested by snapping and unsnapping manually at least twice to determine whether parts of the fastener separate and whether they effect a smooth and secure closure.)	X	
9. LEATHER BUTTONHOLE PIECES		
a. Center section not folded, not rounded or not as specified in 3.3.2		X
10. LEATHER "M" SQUARES		
a. Edges of leather uneven with edges of webbing by 1/8 inch or more		X
11. WIRE BUTTONHOLES		
a. Ends not welded together, i.e., ends or wire piece can be pulled out or disengaged from leather "M" squares		X
b. Bent or malformed, failing to serve intended purpose	X	
12. BUCKLES		
a. Loose, failing to lock suspenders after length adjustment	X	
b. Jammed, failing to effect a smooth length adjustment	X	
c. Reversed, i.e., set to straps with front part of buckles finished on opposite side of cast-off opening		X
13. SHOULDER STRAPS		
a. Improperly assembled, i.e., right shoulder strap finished on top of left shoulder strap		X

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TABLE I. TESTING OF COMPONENTS

Component Characteristic	Test Method	Requirement Applicable to: Individual Unit of Product	Requirement Applicable to: Lot Average	Determinations Per Sample Unit	Inspection Level 1/
BUCKLE (SEE 3.2.4.1)					
Front					
Thickness	gage	X		1	S-1
Material identification	standard commercial		X	2	
Condition and temper 2/	standard commercial				
Back					
Material identification	standard commercial		X	2	
Thickness	gage	X		1	S-1
WIRE BUTTONHOLES (SEE 3.2.4.2)					
Thickness	gage	X		1	S-1
Material identification	standard commercial		X	2	
CAST-OFF ASSEMBLY (SEE 3.2.4.3)					
Wire loop					
Thickness	gage	X		1	S-1
Material identification	standard commercial		X	2	
Suspender hook					
Thickness	gage	X		1	S-1
Material identification	standard commercial		X	2	
Condition and temper 2/	standard commercial				
ALL METAL PARTS					
Finish (see 3.2.4)	standard commercial		X	1	
Corrosion (see 3.5.1)	see 4.4.1		X	1	

1/ The acceptable quality level will be 6.5 dhu for all defects.
2/ A certificate of compliance will be acceptable for this requirement.

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4.2.2.2 Examination of dimensions. Defects pertaining to dimensions shall be classified as follows:

Examine	Defect
Back part of suspenders (8-1/4 inches, plus 1/2 inch, minus 0) from end of wire buttonholes to center line of both shoulder straps (see Figure 1)	not within specified tolerances
Leather buttonholes	under or over specified tolerance
	buttonholes less than 3/4 inch or more than 1 inch long or cut less than 1/2 inch from ends of leather
Leather "M" squares	under or over specified tolerance
Angle of shoulder straps (45 degrees, see Figure 1)	less than 40 or more than 50 degrees.

4.2.2.3 Inspection levels and acceptable quality levels. The inspection levels shall be II of MIL-STD-105 for 4.2.2.1 and S-3 of MIL-STD-105 for 4.2.2.2. The acceptable quality level for 4.2.2.1 shall be 2.5 defects per 100 units (dhu) for major defects and 10.0 dhu for total defects. The acceptable quality level for 4.2.2.2 shall be 6.5 dhu for defects (one class).

4.3 Examination of preparation for delivery. An examination shall be made to determine if packaging, packing, and marking requirements have been met. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 of MIL-STD-105, and the acceptable quality level shall be 2.5 dhu. The shipping container shall be examined for the following defects:

Examine	Defect
Markings (exterior and interior)	Incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any nonconforming component; or component missing, damaged, or otherwise defective affecting serviceability
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, or inadequate stapling
	Bulging or distortion of containers
Content	Number per container is more or less than required or gross or net weight exceeds requirement.

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4.4 Test methods.

4.4.1 Salt spray. Five samples of each metal part shall be subjected to salt spray for 100 hours in accordance with method 811 of FED-STD-151. Corroded areas larger than 1/16 inch in diameter shall be cause for rejection. A corroded area shall be defined as exposure or corrosion of the base metal.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Twelve pairs of suspenders shall be stacked on top of each other and tied with cotton twine or tape. (See 6.2)

5.2 Packing. (See 6.2)

5.2.1 Level A. Six level B packs shall be overpacked packed two in length, one in width, and three in depth in a fiberboard shipping container conforming to type CF or SF, class weather-resistant, grade V3c or V3s, style FOL of PPP-B-636. The inside dimensions of the shipping container shall be approximately 47 inches in length, 16 inches in width and 22 inches in depth. The requirements for closure and strapping shall conform to appendix of PPP-B-636, except that the joints of the boxes shall be sealed by applying 3-inch width cloth tape conforming to class 1, olive-drab color No. X24087 or X34087 of PPP-T-60.

5.2.2 Level B. Sixty pairs of suspenders packaged as specified in 5.1.1 shall be packed in a shipping container as follows: Fold each bundle in half. Place four bundles parallel to the sides of the container and against one end, and place one bundle across the container parallel to the opposite end. The top and bottom of each pack shall be completely covered with a sheet of paper not less than 30 pound basis weight (24 by 36 - 500) conforming to grade B of UU-P-268. The fiberboard shipping container shall conform to type CF or SF, class domestic, style OSC of PPP-B-636 except that the inner flaps shall meet in the center of the box and the outer flaps shall overlap as may (no flap cutting required). The minimum bursting strength of the fiberboard shall be 350 pounds per square inch. The inside dimensions of the container shall be approximately 23-1/2 inches in length, 15-1/2 inches in width, and 6-3/4 inches in depth. The closure of the top and the bottom of the container shall be effected by applying a suitable adhesive (glue) throughout the entire area of contact between the flaps or by a combination of applying a suitable adhesive (glue) to the top flaps and by stapling the bottom flaps.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The suspenders covered by this specification are intended for use by flying personnel.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

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c. First article - Whether preproduction sample is required (see 3.1.1).

d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 Samples. For access to sample of the end item, and color shade samples, address inquiry to the procuring officer issuing the invitation for bids.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 82

Review Activity:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 8440-F098

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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